

Represent and Solve Equations

mathNexa

Name:

Date:

Success Criteria:

- Solve one-step equations using inverse operations
- Write equations and inequalities from situations
- Graph and interpret inequalities accurately

1. A balance scale shows a bag of apples weighing x pounds and a 3-pound weight balancing an 8-pound weight on the other side ($x + 3 = 8$). Which value of x makes the scale balance correctly?



A $x = 24$

B $x = 11$

C $x = 3$

D $x = 5$

2. After using a coupon for 6 dollars off, Leo paid 19 dollars for a board game. What is the original price p in dollars after solving the equation $p - 6 = 19$?

Answer: _____

3. A shipping pallet holds 4 identical sealed boxes with a combined total weight of 48 kg. Which equation represents this situation and gives the weight w of one box?



A $w - 4 = 48$

B $4w = 48$

C $4 + w = 48$

D $\frac{w}{4} = 48$

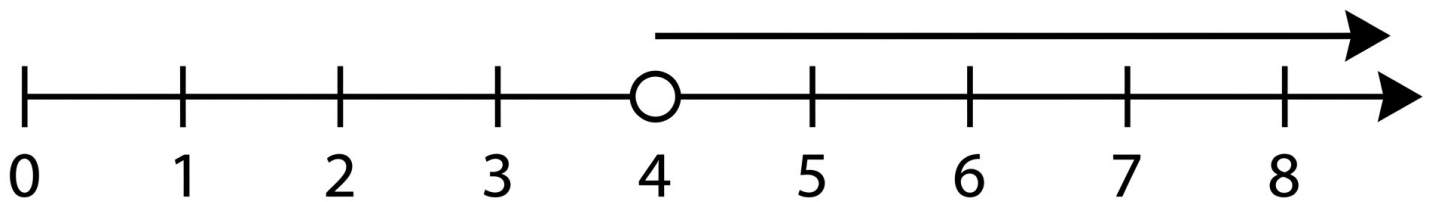
4. A group of 5 friends evenly split the cost c of a shared meal, and each friend paid 14.50 dollars. What is the total meal cost c in dollars found by solving $\frac{c}{5} = 14.50$?

Answer: _____

5. An elevator has a posted safety sign stating that the total load L must be at most 1,200 kg. Which inequality correctly models this condition?



- A $L \leq 1,200$ B $L \geq 1,200$ C $L < 1,200$ D $L > 1,200$
6. A runner completes 0.75 of a marathon course, which equals 18 miles. What is the total length of the marathon m in miles obtained by solving $0.75m = 18$?
- Answer: $m =$ _____ miles
7. Which inequality is represented on a number line that has an open circle at 4 and an arrow pointing to the right toward greater values?



- A $x > 4$ B $x \geq 4$ C $x \leq 4$ D $x < 4$
8. The temperature T inside a laboratory freezer must remain strictly below -12°C ($T < -12$). Does -12°C satisfy the inequality, and what is the highest integer temperature that satisfies it?

Answer: _____

9. Marcus needs to save at least 150 dollars for a science camp. He already has 45 dollars saved and plans to save 15 dollars per week for w weeks. Which inequality represents the number of weeks w he needs to save?

☐ A $45 + 15w \leq 150$

☐ B $15w - 45 \geq 150$

☐ C $45w + 15 > 150$

☐ D $45 + 15w \geq 150$

10. A bakery requires more than 35 kg of flour for a bake. They already have 11 kg in stock and buy additional bags of 6 kg each ($11 + 6b > 35$). What is the minimum whole number of bags b they must purchase?

Answer: $b =$ _____ bags

Answers

1. D

2. 25

3. B

4. 72.50

5. A

6. 24

7. A

8. -13°C

9. D

10. 5